Original Articles

The Diabetes Education Study: A Controlled Trial of the Effects of Diabetes Patient Education

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The Diabetes Education Study (DIABEDS) was a randomized, controlled trial of the effects of patient and physician education. This article describes a systematic education program for diabetes patients and its effects on patient knowledge, skills, self-care behaviors, and relevant physiologic outcomes. The original sample consisted of 532 diabetes patients from the general medicine clinic at an urban medical center. Patients were predominantly elderly, black women with non-insulin-dependent diabetes mellitus of long duration. Patients randomly assigned to experimental groups (N = 263) were offered up to seven modules of patient education. Each content area module contained didactic instruction (lecture, discussion, audio-visual presentation), skill exercises (demonstration, practice, feedback), and behavioral modification techniques (goal setting, contracting, regular follow-up). Two hundred seventy-five patients remained in the study throughout baseline, intervention, and postintervention periods (August 1978 to July 1982). Despite the requirement that patients demonstrate mastery of educational objectives for each module, postintervention assessment 11–14 mo after instruction showed only rare differences between experimental and control patients in diabetes knowledge. However, statistically significant group differences in self-care skills and compliance behaviors were relatively more numerous. Experimental group patients experienced significantly greater reductions in fasting blood glucose (−27.5 mg/dl versus −2.8 mg/dl, $P < 0.05$) and glycosylated hemoglobin (−0.43% versus +0.35%, $P < 0.05$) as compared with control subjects. Patient education also had similar effects on body weight, blood pressure, and serum creatinine. Continued follow-up is planned for DIABEDS patients to determine the longevity of effects and subsequent impact on emergency room visits and hospitalization. DIABETES CARE 1986; 9:1–10.

There is an increasing amount of evidence to suggest that patient education for people with a chronic disease is an essential component of effective disease management. Three comprehensive reviews of the patient education literature[1–3] converge on two general findings. First, any education is better than none, i.e., education in any form (e.g., pamphlets, films, lectures, behavioral modification techniques) is more likely to produce improved regimen compliance and physiologic outcomes than is routine chronic care without formal patient education. The second general finding is that all types of patient education programs are not equal. Two reviews[1,3] looked at the differential effectiveness of didactic and behaviorally oriented patient education. They found behaviorally oriented patient education to be 150–300% more potent than didactic programs. The critical features of behaviorally oriented programs were: (1) emphasis on self-care behaviors rather than the disease process and (2) mechanisms to alter the self-care environment. Some of the mechanisms tested successfully have been reminder systems, daily self-care rituals, regular contact with the same health care provider, and incentives.[3]

While the above findings appear to be true for patients with chronic diseases in general, we are less certain that they apply to diabetes mellitus specifically. In fact, only one true experiment in diabetes patient education was included in the aforementioned reviews. Bowen and associates[4] tested a five-part lecture/demonstration/discussion program. The program produced a modest reduction in body weight in obese patients. However, blood glucose control was not improved as a result of instruction. More recently, Korhonen and associates[5] found no significant improvement in glycemic control resulting from an "intensive" patient education program compared with a control group of patients receiving only printed materials. However, this study has been criticized as having inadequately conceptualized instruction.[6,7] Moreover, the design of this study was such that all instruction was given in an inpatient setting. Both intensive and traditional instruction were tested insofar as they added to glycemic control achieved through hospitalization.

To date, the scientific rationale for patient education as a part of effective diabetes care must be based on studies conducted with patients having other chronic diseases. This is the case not because well-planned diabetes patient education has been tried and found wanting but because the experimental literature on diabetes patient education is too limited to warrant conclusions.

THE DIABETES EDUCATION STUDY

In 1978, the Diabetes Research and Training Center (DRTC) at the Indiana University School of Medicine began its Diabetes Education Study (DIABEDS), a randomized, controlled trial of patient and physician education in the management of diabetes mellitus. From the perspective of patient education, our goal was to determine whether didactic and behavioral methods could be combined into a systematic comprehensive education program for patients with diabetes. The purpose of this article is to describe DIABEDS education for patients and its capacity to improve a variety of cognitive, behavioral, and physiologic outcomes for periods of up to 14 mo. The specific hypotheses tested were as follows: Patients receiving a systematic education program in self-management of diabetes mellitus will be more likely than their control group peers to (1) be knowledgeable about diabetes mellitus, (2) possess skills necessary to comply with therapeutic regimens, (3) comply with self-care recommendations, (4) control glucose homeostasis, and (5) decrease the risks of chronic complications.

TABLE 1
Summary of baseline assessment of 532 DIABEDS patients

Demographics	
Age median	58.1 yr
Sex (% Women)	79%
Race (% Black)	72%
Duration median	6.3 yr
Treatment modalities	
Diet alone	16%
Diet plus insulin	70%
Diet plus oral hypoglycemic agents	14%
Plasma glucose (mean ± SD)	
Fasting	217 ± 92 mg/dl
2-h postprandial	332 ± 111 mg/dl
Glycosylated hemoglobin (mean ± SD)	10.7 ± 3.1%
Weight status (mean ± SD)	
Body weight	84 ± 20 kg
Percentage of ideal body weight	142 ± 31%
Concurrent hypertension	67%
Microvascular complications	
Background retinopathy	42%
Proliferative retinopathy	8%
Proteinuria	9%
Serum creatinine (mean ± SD)	1.10 ± 0.38 mg/dl
Peripheral neuropathy	51%

METHODS

Subjects. Patients in DIABEDS were recruited from the diabetic patient population in a general medicine clinic at the Indiana University Medical Center. Criteria for patient inclusion in the study were that: (1) their clinic record contain either two fasting blood glucose (FBG) values >130 mg/dl, one FBG >150 mg/dl, or a 2-hr postprandial blood glucose value >250 mg/dl; (2) they be able to perform at least two basic self-care tasks (e.g., insulin administration, urine glucose and ketone testing, food selection); (3) they suffer no concurrent psychiatric or terminal illness; (4) their primary care provider be an internal medicine resident; and (5) they give informed consent.

Between August 1978 and February 1981, 532 subjects were recruited into DIABEDS from a clinic population of approximately 1800 diabetic patients. The results of an independent DRTC assessment of this sample are presented in Table 1. Consistent with recent epidemiologic profiles of diabetes, this sample was largely elderly, female, and Black. Ninety-five percent of the patient sample had non-insulin-dependent diabetes mellitus, as evidenced by elevated postprandial C-peptide levels. The large majority of patients were obese, yet only 16% were managed by diet alone. Two-thirds of the sample had hypertension. Nonproliferative retinopathy and peripheral neuropathy were the most common chronic complications of diabetes, as each was observed in about half the sample. In contrast, renal disease was relatively rare.

Clinical setting. Patients in DIABEDS were cared for in the general medicine clinic (GMC) of a university-affiliated city/county hospital. Eighty-six internal medicine residents are assigned to staff the GMC one half-day per week with each resident responsible for the same half-day throughout his/her residency. Each resident assumes responsibility for a panel of approximately 75 patients, 5–8 of whom have diabetes mellitus.

Nine weekly half-day clinics are organized into three teams of health care providers. Each team consists of three or four residents (at different postgraduate levels), a nurse practitioner, a registered nurse, and a senior staff physician from the hospital department of medicine. A nearby diet clinic serves as a referral facility for the GMC.

The GMC had no individual serving in the role of a diabetes patient educator. The nursing staff dispensed written patient education materials, taught insulin injection technique, and reinforced physicians' orders regarding compliance with therapeutic regimens. However, neither formal diabetes

TABLE 2
Incomplete-blocks design for assignment of patients to treatment conditions in DIABEDS

	Team assignment to groups			
Senior staff physicians	Control (group 1)	Patient education (group 2)	Physician education (group 3)	Physician and patient education (group 4)
A	A_1	A_2	A_3	—
B	B_1	B_2	—	B_3
C	C_1	—	C_2	C_3
D	—	D_1	D_2	D_3
E	E_1	E_2	E_3	—
F	F_1	F_2	—	F_3
G	G_1	—	G_2	G_3
H	—	H_1	H_2	H_3
I	I_1	—	I_2	I_3
Teams	7	6	7	7
Residents	21	20	23	22
Patients	135	125	134	138

information programs, clinic-approved protocols for compliance behavior modification, nor guidelines for follow-up and reinforcement were a part of routine diabetes care in the GMC throughout the study.

Experimental design. Three teams of residents cared for DIABEDS patients during each of the nine half-day clinics in the GMC. Therefore, the 86 residents were grouped a priori into 27 clinic teams. Six senior staff physicians and three pairs of staff members each bore attending responsibilities for three clinic teams throughout the week's schedule. Because the GMC structure and pattern of staff responsibilities could not be reconciled evenly with assignment into four DIABEDS groups, a randomized, incomplete-blocks design was adopted. Table 2 contains a representation of assignment to groups in DIABEDS. An incomplete-blocks pattern was designed such that a senior staff physician's teams would be assigned to three of the four experimental groups. The "empty" group for each assignment row (rows A–I in Table 2) was varied systematically. Senior staff physicians were assigned randomly to rows. Their GMC teams and therefore patients receiving care from residents in those teams were then assigned randomly to the three designated groups. In this fashion, the potential influences of senior staff, postgraduate level of residents, nursing staff, day of the week, and time of day were evenly dispersed throughout the DIABEDS design. Risk of contamination was minimized because all residents and patients for a given team were assigned to the same DIABEDS condition.

Group 1, the control group, was the routine education condition. Residents received instruction about diabetes only insofar as it was routinely available in the programs of this university-affiliated teaching hospital. Patients received diabetes education only as it was offered as a routine part of care in the GMC. Patients in group 2 received a systematic program in diabetes education from staff of the DRTC but group 2 physicians received no special instruction. Group 3 physicians received an educational program on ambulatory diabetes management from DRTC faculty diabetologists but their patients received only routinely available diabetes education. Finally, in group 4 both physicians and their patients received DRTC-initiated instruction.

Since the purpose of this article is to describe the effects of patient education, we will refer from this point on to experimental and control group patients in such a way that experimental group patients include those assigned randomly to DIABEDS groups 2 and 4. Control group patients were those assigned to groups 1 and 3.

Instrumentation and assessment. All DIABEDS patients received an independent day-long assessment from DRTC personnel. Elements of the assessment included a nurse-administered patient history, a dietitian-administered nutritional history, an examination of physical characteristics and organ systems related to diabetes, e.g., body weight, blood pressure, eyes, nerves, feet, and laboratory measurements of glucose homeostasis, renal function, and cardiac status.

Measures of patient knowledge, skills, and compliance were derived from responses to the two "histories" administered by DRTC nurses and dietitians. These responses were used to diagnose the educational needs of patients. A test–retest procedure performed for 40 DIABEDS patients showed that ≥89% of content area diagnoses made at the time of the first administration were replicated 3 mo later.

Physiologic data used to characterize glucose homeostasis in this study include FBG, glycosylated hemoglobin (HbA_1) and body weight. The assay system for HbA_1 (Isolab columns, Quik-Sep method with Aldinine Eliminator, Isolab Incorporated, Akron, Ohio) measured only the stable fraction.[8] Systolic and diastolic blood pressure and serum creatinine were studied to represent risk factors for vascular complications of diabetes.

TABLE 3
Safety-level objectives for DIABEDS patient education

Understanding diabetes
Defines diabetes as "too much sugar in the blood"
Describes mode(s) of control as diet, plus medication and exercise
Wears diabetes identification at all times
Acute complications
Describes at least two symptoms of hypoglycemia and steps in treatment
Carries an immediate source of carbohydrate
Describes at least two symptoms of hyperglycemia and at least two conditions that can result in hyperglycemia
Verbalizes need to call health care professional if symptoms of hyperglycemia or hypoglycemia remain or worsen, and/or if ill for >24 h
Diabetic and hypertensive medication
Identifies purpose of prescribed medication
States proper time(s) and doses
Mixes insulin thoroughly (modified insulins)
Fills syringe without contamination
Draws up accurate dose (±2U)
Diet and activity
Uses exchange lists to solve problems of food substitution
Selects and prepares amounts/portions of foods as recommended
Foot care
Verbalizes need to contact health care professional regarding nonhealing foot lesions
States one specific result of poor foot care
Describes procedure for foot care
Urine test
Tests for glycosuria according to directions
Tests for ketonuria according to directions
Verbalizes need to call health care professional if urine (3–4+) × 3 with ketones

The DIABEDS patient assessment was made during baseline and postintervention periods. Patient recruitment and baseline assessment began in August 1978 and continued until February 1981. Postintervention assessment began in February 1981 and continued through July 1982. No experimental group patients were reassessed sooner than 6 mo after completion of nondietary DIABEDS instruction and no sooner than 2 mo after dietary intervention. Experimental patients eligible for reassessment were paired with control patients on the basis of time of baseline assessment and then scheduled for postintervention evaluation. Due to staffing limitations, nonlaboratory assessments could not be conducted by personnel who were blind to subjects' experimental condition. However, in no case was an experimental group patient reassessed by a nurse or dietitian who had been their teacher.

DIABEDS patient education. Patient education in DIABEDS was designed to ensure that patients satisfy a minimal set of safety-level objectives regarding knowledge, self-management skills, and compliance with diabetes regimens. A list of those objectives appears in Table 3. The safety-level objectives were selected by a multidisciplinary group of health care professionals, who evaluated an exhaustive set of objectives with respect to clinical importance and amenability to improvement through instruction. The adopted set of objectives covered seven content areas of patient education: understanding diabetes, acute complications, antidiabetic medication, antihypertension medication, diet and activity, foot care, and urine testing.

Questions keyed to the objectives were embedded into the history components of the baseline assessment. For patients who were to receive education (groups 2 and 4), responses to the germane history questions were judged for appropriateness according to seven protocols for educational diagnosis. Each protocol covered objectives in one content area. Patients were enrolled in the appropriate module(s) of instruction depending on the educational needs diagnosed according to the protocols. Table 4 contains the results of educational diagnoses for the 263 patients assigned to groups 2 and 4, as well as the number of patients subsequently completing the various modules.

Each educational module contained three elements. The first was didactic instruction in which lecture, discussion, and/or audiovisual materials conveyed necessary information. DRTC instructors, i.e., nurses and dietitians, elicited comprehension periodically throughout the lesson. Also, safety-level skills were taught formally by means of demonstration, return demonstration, and feedback. Information and skills teaching was repeated until mastery of the particular objective was demonstrated by patients. The second element was a goal-setting exercise. At the end of classroom instruction patients negotiated with their instructor the compliance goals that they believed they could attain. These goals were incorporated into a contract that was signed by both patient and instructors. Copies were retained by each. Finally, each module of instruction contained a reinforcement schedule in which the patient was contacted by phone 2 and 6 wk after instruc-

TABLE 4
Number of DIABEDS patients diagnosed as needing and completing instruction

Module	No. patients needing instruction*	No. patients completing instruction
Understanding diabetes	200	129
Acute complications		
Hyperglycemia	183	147
Hypoglycemia	172	147
Diabetic medications	143	96
Antihypertensive medications	131	83
Diet and activity	200†	86
Foot care	208	139
Urine testing	206	141

*Two hundred sixty-three patients were assigned to experimental group.
†Two hundred eight of 263 experimental group patients received nutritional assessment.

tion. Progress toward compliance goals was assessed at each contact and reinforced where appropriate. Renegotiation of therapeutically more effective goals took place when patients succeeded at attaining intermediate goals.

Patients, on average, received 2.4 modules of instruction. Except for the diet and activity module, classroom contact time was ≈90 min per module. Those patients receiving diet instruction interacted with DRTC dietitians for ≈5 h over the course of 8 wk. In addition to the patient education elements mentioned earlier, patients in diet instruction also received personalized computer-generated meal plans and menus[9], as well as a home visit. All nondietary patient education in DIABEDS took place in the period of October 1979 to July 1981. Diet education was given between July 1980 and February 1982.

A total of 119 of the 135 (88%) reassessed experimental group patients received at least one module of instruction. In contrast, 103 of the 140 (74%) reassessed control group patients reported receiving some diabetes instruction since baseline assessment. It would appear that both experimental and control patients had ample opportunity to receive self-care instruction. The crucial difference between groups therefore is the systematic nature of DIABEDS instructions compared with routine education described in CLINICAL SETTING.

Data analysis. Because DIABEDS patient education, like any similar program, relied ultimately on the voluntary cooperation of patients, all reassessed patients (whether they received instruction, reported to receive instruction, or not), were included in data analyses. In this fashion, we intended to test the effects of a systematic program under the most realistic conditions, thus increasing our capacity to generalize to actual care settings.

Responses from the postintervention patient history interview were coded to reflect whether knowledge, skills, and compliance conformed to the safety-level objectives for nondietary modules. The number and percentage of patients in the experimental and control groups who satisfied the objectives was calculated. Fisher's exact probability test was then used to obtain the difference between experimental and control group percentages that could have occurred by chance.

The nutritional assessment contained multi-item measures of knowledge of exchange lists and of weighing and portioning skills. In addition, each patient responded to a 24-h recall interview concerning food intake, responses to which were coded to indicate caloric intake vis-à-vis the diet prescription, i.e., compliance. Scores on these measures of dietary knowledge, skill, and compliance were analyzed using analyses of covariance. In this model, differences between the original four groups (overall F-test) and an a priori linear contrast of groups 1 and 3 versus groups 2 and 4 (t-test) were conducted on adjusted postintervention data. Data were adjusted to remove variation that could have been predicted on the basis of baseline status.[10]

Physiologic data, i.e., FBG, HbA_1, body weight, blood pressure, and serum creatinine) were likewise analyzed using parametric statistical models. As will be described in RESULTS, randomization in the DIABEDS design did not produce equivalent groups at baseline with regard to glucose homeostasis. Group 1 patients were in relatively better control and weighed less, while group 4 patients were relatively more hyperglycemic and heavier. Both groups 1 and 4 had higher creatinine values than did groups 2 and 3. The analytic model of choice in this situation is one-way analysis of covariance on postintervention values adusted for baseline covariation. An assumption of this model is that the amount of adjustment in each group is similar. When the assumptions of "parallel within-group regressions" are not met, the analysis is of doubtful validity. This assumption was satisfied in the analyses of covariance for body weight and blood pressure; the BMDP1V statistical program[11] was used to obtain this measurement. However, significant nonparallelism was found in the covariance analyses for FBG, HbA_1, and serum creatinine.

In the cases of FBG, HbA_1, and creatinine, one-way analyses of variance were conducted on change values, i.e., postintervention minus baseline. As with the previous model, a test of the appropriate linear contrast was conducted to evaluate the effect of DIABEDS patient education. An assumption of this model is that within-group variation in change scores is similar. The assumption was met for the analysis of FBG, so the t-test associated with the contrast was based on a pooled-variance estimate. A separate-variance estimate or "t-prime" test was required for the HbA_1 and creatinine analyses, however, as heterogeneity of within-group variation of changes was found. Statistical Package for the Social Sciences (SPSS) program ONE-WAY was used to conduct analyses of variance on FBG, HbA_1, and creatinine change values.[12]

The a priori contrasts conducted to represent main effects for patient education were each accompanied by contrasts to test the main effect of physician education (groups 1 and 2 versus groups 3 and 4) and the interaction of patient and physician education. While the effects of physician education are beyond the scope of this article, it is important to note that there were no significant interaction effects; thus, the patient education effects reported herein are independent of

TABLE 5
Percentage of patients who satisfied knowledge objectives for diabetes patient education at postintervention assessment

Objective	DIABEDS groups: Experimental (groups 2 and 4) %	N	DIABEDS groups: Control (groups 1 and 3) %	N	Fisher's exact test*
Defines diabetes	41.4	133	38.1	139	0.3375
List modes of control	99.2	133	99.3	139	0.7618
States purpose of diet	97.7	131	96.4	138	0.3907
States name, dosage, and purpose of insulin/oral agent	57.4	115	58.0	119	0.5887
Lists at least two symptoms of hyperglycemia	47.4	133	50.4	139	0.7306
Lists at least two causes of hyperglycemia	51.1	133	30.9	139	0.0005
Lists at least two symptoms of hypoglycemia	79.8	114	73.1	119	0.1463
Describes hypoglycemia treatment	89.5	114	86.6	119	0.3150
Knows urine test conditions that indicate need for health care professional	92.5	133	78.8	137	0.0011

*One-tailed probability in the direction of the experimental group's having the higher percentage.

and essentially additive with respect to effects of DIABEDS physician education. Also, it should be noted here that preliminary "general-linear-model" analyses of covariance showed that neither senior staff physicians nor resident level of training had significant main effects on any of the reported variables.

RESULTS

During the period of postintervention assessment, 275 DIABEDS patients (135 experimental and 140 control) agreed to submit to all or part of a second series of laboratory, history, and physical evaluations. The median interval between instruction and reassessment was 6.3 mo for the diet module and 11.8–14.3 mo for all other patient education modules. Every reasonable effort was made to accommodate the entire patient sample, e.g., payment for transportation costs, scheduling on weekends. Reasons for attrition were death, 30 patients; physical or psychological incapacitation, 43; transfer to a senior staff physician, 32; relocation, 13; work conflict, 24; personal reasons, 45; multiple failure to keep DRTC appointments, 11; lost contact by phone and mail, 58.

The rate of attrition was relatively uniform in all DIABEDS groups. Dropouts were not significantly different overall from reassessed patients with respect to age, duration of diabetes, FBG, and body weight. However, there was differential attrition within groups regarding HbA_1 values. Reassessed patients had lower HbA_1 values than did dropouts in group 1 (10.18% versus 11.36%, $P < 0.05$) and in group 3 (10.44% versus 11.47%, $P < 0.10$). There were no such differences between dropouts and reassessed patients in the experimental groups (2 and 4). This pattern of attrition does not significantly alter our interpretation of most results. The interpretation of HbA_1 data, however, should be tempered by the fact that control patients in poorer control tended to decline reassessment, thus making it more difficult to demonstrate a significant effect for patient education.

The remaining sample still allowed powerful analysis of educational effects. For dichotomous variables, i.e., achievement of knowledge, skill, and self-care objectives, significant differences (one-tailed $P < 0.05$) between experimental and control group percentages of 4.3–13.0% were detectable, depending on whether the percentages were located at the extremes or toward the middle of the 0–100% range, respectively. Significant differences (one-tailed $P < 0.05$) between groups on continuous (physiologic) variables of a magnitude of about two-tenths of the within-group standard deviation also were detectable.

Effect on patients' knowledge. Table 5 contains the percentages of patients in the experimental and control groups who satisfied knowledge objectives for the patient education modules during the reassessment. More than 85% of patients in both groups could list all modes of blood glucose control, state the purpose of a diabetes diet, and describe treatment for hypoglycemia. There was a significant difference in the groups' ability to list causes of hyperglycemia (experimental, 51.1%; control, 30.9%; Fisher's exact $P < 0.001$) and in their knowledge of urine test conditions that indicate the need for a health care professional (experimental, 92.5%; control, 78.8%; $P < 0.005$). A finding not presented in Table 5 was a significant increase in patient knowledge about

TABLE 6
Percentage of patients who satisfied skill objectives for diabetes patient education at postintervention assessment

Objective	DIABEDS groups: Experimental (groups 2 and 4) %	N	Control (groups 1 and 3) %	N	Fisher's exact test*
Draws insulin dose to within 2U of prescription	90.2	92	90.0	100	0.5767
Urine testing skills					
Wets product correctly	79.5	132	78.5	130	0.4746
Waits appropriate time	47.7	132	30.8	130	0.0036
Uses color chart	95.5	132	91.5	130	0.1502
Interprets correct result	93.2	132	82.3	130	0.0058

*One-tailed probability in the direction of the experimental group's having the higher percentage.

the diabetes exchange list system. Experimental patients scored an average of 73.3% on a 50-item written test of exchanges, compared with 70.2% (t = 2.937, df = 240, P < 0.005). Any other group differences in knowledge that may have occurred as a result of experimental patients' mastering of the objectives of patient education modules were not sustained over the 11–14-mo period.

Effects on patients' skills. Table 6 contains percentages of patients in the experimental and control groups who satisfied skill objectives for the antidiabetic medication and urine testing modules. Ninety percent of insulin-taking patients in both groups were able to draw the prescribed insulin dosage to within a 2-U tolerance. Four urine-testing skills were evaluated for patients who tested their urine regularly: wetting the product, waiting the appropriate time, using the color chart, and intepreting results. Product wetting and color chart usage were unchanged but similarly prevalent skills in the two groups. Patients receiving urine testing instruction were more likely to wait the appropriate time before reading (47.7% in the experimental group versus 30.8% in the control group, P < 0.005) and to interpret the semiquantitative glucose result accurately (93.2% versus 82.3%, experimental versus control, respectively, P < 0.01).

Patients were asked to perform five standardized tasks, requiring food portioning and/or weighing, according to the requirements of diabetes exchanges. At baseline, both groups on average performed 3.2 tasks correctly. At reassessment, experimental patients averaged 3.7 tasks correct versus 3.0 tasks correct for control patients (t = 4.704, df = 250, P < 0.001).

Effects on patient self-care behavior. All subjects who received patient education were asked to carry some sort of diabetes identification. At reassessment, 82.7% of experimental patients were able to produce legible identification. Among control group patients, 66.2% were able to produce a diabetes identification (Fisher's exact P < 0.005). In addition, patients on insulin were instructed about the reasons for carrying a quickly metabolized source of carbohydrate. During reassessment, 63.2% of insulin-taking experimental patients were able to produce an acceptable source. Only 47.9% of control patients were able to do so (P < 0.05). Patient self-report revealed that 85.1% of experimental patients always complied with their insulin or oral agent regimen. The comparable percentage in the control group was 79.8%. This difference was not statistically significant.

Dietary recall over a 24-h period was used as a measure of compliance with the dietary regimen. Caloric intake over the previous 24 h was estimated by DRTC dietitians using the methods of the U.S. Health and Nutrition Survey.[13] The absolute value of the difference between 24-h caloric intake and the prescribed caloric limit was calculated for each patient. The average discrepancy for experimental patients was 427 kcal, whereas the average control group discrepancy was 530 kcal. This apparent 100-kcal-per-day difference was statistically significant (t = −2.485, df = 248, P < 0.05).

Effects on glucose homeostasis. To characterize the effects of DIABEDS patient education on glucose homeostasis, we analyzed changes in FBG, HbA_1, and body weight. Of 275 patients volunteering for all or part of postintervention assessment, 247 allowed DRTC personnel to draw blood samples. Of these patients, 238 presented in a fasted state.

Table 7 contains the means for experimental (groups 2 and 4) and control (groups 1 and 3) patients of FBG at baseline, postintervention, and the change over time (postintervention minus baseline). The overall F-ratio in the analysis of variance testing the equality of change in the four original groups was 2.034 (df = 3,234; P = 0.110). The a priori linear contrast to test the significance of change in groups 2 and 4 versus groups 1 and 3 showed a statistically significant effect for patient education (t = 1.850, df = 234, one-tailed P < 0.05). The significance of this test reflects the mean change within experimental groups (a 27.5-mg/dl drop in FBG) compared with the corresponding reduction among control patients between baseline and postintervention assessments (2.8 mg/dl).

Mean values for baseline, postintervention, and change in HbA_1 also are presented in Table 7. The overall F-ratio in the analysis of variance on change values indicated significant differences among groups (F= 3.119; df = 3,243; P < 0.05). However, heterogeneity of within-group variance of changes

TABLE 7
Experimental and control group means for fasting blood glucose (FBG), glycosylated hemoglobin $(HbA)_1$, and body weight

	DIABEDS groups						
	Experimental			Control			
	2		4	1		3	Total
Number of patients	61		54	65		58	238
Baseline FBG (mg/dl)	213.8		229.2	201.1		209.6	212.8
Postintervention FBG (mg/dl)	197.7		190.2	208.7		196.5	198.8
FBG change (mg/dl)*	−15.9		−39.0	+7.6		−13.1	−14.0
Combined mean change (mg/dl)†		−27.5			−2.8		
Number of patients	64		56	67		60	247
Baseline HbA_1 (%)	10.17		11.34	10.19		10.51	10.52
Postintervention HbA_1 (%)	10.23		10.42	10.74		10.65	10.51
HbA_1 change (%)	+0.06		−0.92	+0.56		+0.14	−0.01
Combined mean change (%)†		−0.43			+0.35		
Number of patients	66		53	65		51	245
Baseline body weight (kg)	84.63		87.89	84.04		85.65	85.44
Postintervention body weight (kg)	83.02		85.77	84.54		84.08	84.28
Adjusted-postintervention weight (kg)	83.79		83.42	85.87		83.88	
Combined mean (kg)†		83.61			84.87		

*Postintervention minus baseline.
†Unweighted means are reported under the assumption that differential attrition was random and does not represent a result of treatments.

was found ($P < 0.001$); so the t-prime test of the linear contrast was used, based on separate-variance estimates. Patient education was shown to have a significant effect ($t' = 2.214$, df ≈ 185.3, one-tailed $P < 0.05$). The mean reduction HbA_1 values for patients in the experimental group was 0.43%, whereas mean change in control group patients represented a 0.35% increase in HbA_1.

As previously mentioned in DATA ANALYSIS, analysis of covariance was conducted on postintervention body weight values using baseline weight as the covariate. Mean values for baseline, postintervention, and covariate-adjusted postintervention body weight in experimental and control groups are presented in Table 7. The overall F-ratio testing equality of adjusted means was 2.439 (df = 3,240; $P = 0.065$). The linear contrast testing the effect of patient education indicated a statistically significant effect ($t = -1.766$, df = 240, one-tailed $P < 0.05$). The mean of covariate-adjusted body weights in the experimental groups was 83.61 kg. The corresponding value in control patients was 84.87 kg.

Effects on risk factors for chronic complications. The effects of patient education on risk factors for chronic vascular complications were tested in terms of patients' blood pressure (systolic and diastolic) and serum creatinine. Experimental and control group means at baseline and at postintervention assessment are presented in Table 8.

Analyses of covariance were conducted on postintervention systolic and diastolic blood pressure values that were covariate-adjusted for baseline status. The overall F-ratio (2.277) for systolic blood pressure was marginally significant (df = 3,251; $P = 0.08$). The linear contrast of combined experimental and control groups' adjusted means (141.8 mm Hg versus 145.7 mm Hg, respectively) showed a significant effect for patient education ($t = -1.707$, df = 251, one-tailed $P < 0.05$). The effects on diastolic blood pressure were somewhat more dramatic. There were significant postintervention differences among the original four DIABEDS groups ($F = 2.926$, df = 3,251; $P < 0.05$). Specifically, the mean adjusted diastolic blood pressure of patients receiving patient education was significantly less than that of control patients (81.7 mm Hg versus 84.4 mm Hg, respectively, $t = -2.548$, df = 251, one-tailed $P < 0.01$).

As previously mentioned in DATA ANALYSIS, significant nonparallelism of within-group regressions was found in the analysis of covariance for serum creatinine values; thus, an analysis of variance on changes in creatinine (postintervention minus baseline) was conducted. In this analysis, heterogeneity of within-group variances was also found, thus making the separate-variance estimate t'-test of the linear contrast the primary test of the effect of patient education. The contrast showed a significant effect for patient education ($t' = -2.174$, df = 152, one-tailed $P < 0.05$). The significance of this test reflects a 0.05-mg/dl decrease in serum creatinine among experimental patients, compared with a 0.06-mg/dl increase in control patients.

Congruence of physiologic effects. The last series of analyses was conducted to determine the extent to which the uni-

TABLE 8
Experimental and control group means for systolic and diastolic blood pressure (BP) and for serum creatinine

	DIABEDS groups				
	Experimental		Control		
	2	4	1	3	Total
Number of patients	69	58	67	62	256
Baseline systolic BP (mm Hg)	139.9	140.4	137.2	142.5	139.9
Postintervention systolic BP (mm Hg)	138.9	145.0	144.9	146.4	143.7
Adjusted postintervention systolic BP (mm Hg)	138.9	144.7	146.3	145.0	
Combined mean BP (mm Hg)*	141.8		145.7		
Number of patients	69	58	67	62	256
Baseline diastolic BP (mm Hg)	84.7	81.8	81.4	83.1	82.8
Postintervention diastolic BP (mm Hg)	82.4	81.3	85.2	83.4	83.1
Adjusted postintervention diastolic BP (mm Hg)	81.9	81.5	85.5	83.4	
Combined mean BP (mm Hg)*	81.7		84.4		
Number of patients	49	36	51	43	179
Baseline serum creatinine (mg/dl)	1.04	1.19	1.11	1.07	1.10
Postintervention serum creatinine (mg/dl)	1.02	1.11	1.19	1.10	1.11
Creatinine change (mg/dl)†	−0.02	−0.08	+0.08	+0.03	+0.01
Combined mean (mg/dl)*	−0.05		+0.06		

*Unweighted means are reported under the assumption that differential attrition was random and does not represent a result of treatments.
†Postintervention minus baseline.

variate results presented heretofore represent discrete effects or a lesser number of congruent effects. Multivariate analysis of variance (MANOVA) on change from baseline to postintervention assessment was applied using the following variables: FBG, HbA_1, body weight, systolic blood pressure, and diastolic blood pressure. (Serum creatinine was not included due to missing data.) The effects of DIABEDS patient education were found to be essentially congruent on the two variables FBG and HbA_1. Likewise redundant were the effects on systolic and diastolic blood pressure. The effect on body weight was found to be independent of those occurring in the other two clusters of variables.

Furthermore, systematic application of MANOVA to combinations of variables from these three clusters showed that the strongest discrimination between experimental and control group patients could be made on the basis of changes in HbA_1 and diastolic blood pressure values (overall $F = 5.384$, $df = 2{,}222$; $P < 0.01$). Both variables made significant ($P < 0.05$) partial contributions to the MANOVA. When body weight was added to this MANOVA discriminant function, the significance of the overall F was further enhanced ($F = 4.483$, $df = 3{,}221$; $P < 0.005$). However, the partial F for body weight was not statistically significant. Thus, the effects of DIABEDS patient education on physiologic variables appear to cluster into two discrete and significant areas—metabolic control (FBG or HbA_1) and blood pressure (systolic or diastolic). The independent effect on body weight was not sufficiently strong to contribute multivariately to the further discrimination of experimental and control group patients.

DISCUSSION

The Diabetes Education Study was designed in part to determine whether a systematic patient education program could affect for a prolonged period of time patient knowledge, skills, self-care behavior, glucose homeostasis, and risk factors for chronic complications. The program tested was a combination of didactic teaching methodologies (lecture, discussion, audiovisual presentation), skill exercises (demonstration, practice, and feedback), and behavioral modification techniques (goal setting, contracting, regular follow-up).

While we observed a statistically significant but small difference between groups on exchange list knowledge 6 mo after instruction, other knowledge differences observed 11–14 mo after instruction were rare. The lack of differences on knowledge objectives however did not preclude numerous, prolonged, and substantive differences in skills and self-care behaviors, all of which favored patients receiving DIABEDS instruction. Patient education modestly improved metabolic control in diabetes patients. The 27.5-mg/dl decrease in experimental patients' FBG compared with the control group reduction (2.8 mg/dl) represents an effect of approximately one-quarter of the within-group standard deviation. The corresponding difference between experimental and control group HbA_1 changes was 28% of a standard deviation. While these differences are not large in an absolute sense, it should be recalled that control group patients with relatively worse metabolic control at baseline tended to decline reassessment.

Nevertheless, a significant effect for patient education was demonstrated.

The baseline inequities among the DIABEDS groups for FBG, HbA_1, and body weight deserve additional comment as they pertain to interpretation of change in clinical variables. In addition to any potential effect of patient education, random variation and measurement error, i.e., statistical regression to the mean, also may have caused glucose homeostasis to improve in experimental groups. However, baseline means for clinical variables in all groups were indicative of uncontrolled diabetes. Neither random variation nor statistical regression could have been expected to force group means markedly closer together. Also, because of the relative stability inherent in measures of body weight and HbA_1, the potential for regression to the mean appears small in comparison with observed changes in the groups.

Another source of caution stems from the fact that all DIABEDS patients were chronically hyperglycemic and obese. Despite the likelihood that these patients may be representative of diabetic populations in many other settings, patients in this study may have represented a fertile testing ground. Perhaps any amount of patient education was bound to improve matters. By the same token, obese, hyperglycemic non-insulin-dependent patients may represent the other extreme—"tough cases" for whom only heroic efforts will produce a noticeable change in diabetes control. Only further research in this and other settings will determine which is the case.

These caveats notwithstanding, we conclude that systematic education can have a demonstrable, prolonged effect on patient self-care skills and behaviors and on intermediate indicators of glucose homeostasis and chronic vascular complications. We will follow these patients for another year to determine the longevity of observed effects and to assess long-term effects on emergency room and hospital utilization.

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